DDX41_CpG_island_main_hits_TFs

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```r
options(repos = c(CRAN = "https://cloud.r-project.org"))

install.packages("rmarkdown")
```

```
## Installing package into 'C:/Users/Charvi Khanna/AppData/Local/R/win-library/4.6'
## (as 'lib' is unspecified)

## package 'rmarkdown' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
##  C:\Users\Charvi Khanna\AppData\Local\Temp\RtmpOSOXto\downloaded_packages
```

```r
install.packages("knitr")
```

```
## Installing package into 'C:/Users/Charvi Khanna/AppData/Local/R/win-library/4.6'
## (as 'lib' is unspecified)

## package 'knitr' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
##  C:\Users\Charvi Khanna\AppData\Local\Temp\RtmpOSOXto\downloaded_packages
```

```r
install.packages("tinytex")
```

```
## Installing package into 'C:/Users/Charvi Khanna/AppData/Local/R/win-library/4.6'
## (as 'lib' is unspecified)

## package 'tinytex' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
##  C:\Users\Charvi Khanna\AppData\Local\Temp\RtmpOSOXto\downloaded_packages
```

```r
BiocManager::install("TFBSTools")
```

```
## 'getOption("repos")' replaces Bioconductor standard repositories, see
## 'help("repositories", package = "BiocManager")' for details.
## Replacement repositories:
##     CRAN: https://cloud.r-project.org
```

```
## Bioconductor version 3.23 (BiocManager 1.30.27), R 4.6.0 (2026-04-24 ucrt)

## Warning: package(s) not installed when version(s) same as or greater than current; use
##   `force = TRUE` to re-install: 'TFBSTools'

## Warning in .available.both(repos, method, ...): Some listed binary packages
## have no source

## Installation paths not writeable, unable to update packages
##   path: C:/Program Files/R/R-4.6.0/library
##   packages:
##     boot, class, cluster, KernSmooth, lattice, MASS, Matrix, mgcv, nlme, nnet,
##     rpart, spatial, survival

## Old packages: 'askpass', 'BiocGenerics', 'BiocManager', 'bit', 'bit64', 'blob',
##   'broom', 'bslib', 'cachem', 'callr', 'caTools', 'cellranger', 'cli', 'clipr',
##   'conflicted', 'cpp11', 'crayon', 'curl', 'data.table', 'DBI', 'dbplyr',
##   'digest', 'dplyr', 'dtplyr', 'evaluate', 'farver', 'fastmap', 'filelock',
##   'fontawesome', 'forcats', 'formatR', 'fs', 'futile.logger', 'gargle',
##   'generics', 'ggplot2', 'ggseqlogo', 'glue', 'googledrive', 'googlesheets4',
##   'gtable', 'gtools', 'haven', 'highr', 'hms', 'htmltools', 'httr', 'httr2',
##   'ids', 'isoband', 'jquerylib', 'jsonlite', 'lambda.r', 'lifecycle', 'locfit',
##   'lubridate', 'magrittr', 'matrixStats', 'memoise', 'modelr', 'openssl',
##   'openxlsx', 'pillar', 'pkgconfig', 'prettyunits', 'processx', 'progress',
##   'ps', 'purrr', 'R6', 'ragg', 'rappdirs', 'Rcpp', 'RcppArmadillo', 'readr',
##   'readxl', 'rematch', 'rematch2', 'reprex', 'restfulr', 'rlang', 'RSQLite',
##   'rstudioapi', 'rvest', 'S4Vectors', 'S7', 'sass', 'scales', 'selectr',
##   'stringr', 'sys', 'systemfonts', 'textshaping', 'TFMPvalue', 'tibble',
##   'tidyr', 'tidyselect', 'tidyverse', 'timechange', 'tzdb', 'utf8', 'vctrs',
##   'viridisLite', 'vroom', 'withr', 'xfun', 'xml2', 'yaml', 'zip'
```

```r
BiocManager::install("Biostrings")
```

```
## 'getOption("repos")' replaces Bioconductor standard repositories, see
## 'help("repositories", package = "BiocManager")' for details.
## Replacement repositories:
##     CRAN: https://cloud.r-project.org

## Bioconductor version 3.23 (BiocManager 1.30.27), R 4.6.0 (2026-04-24 ucrt)

## Warning: package(s) not installed when version(s) same as or greater than current; use
##   `force = TRUE` to re-install: 'Biostrings'
## Warning: Some listed binary packages have no source

## Installation paths not writeable, unable to update packages
##   path: C:/Program Files/R/R-4.6.0/library
##   packages:
##     boot, class, cluster, KernSmooth, lattice, MASS, Matrix, mgcv, nlme, nnet,
##     rpart, spatial, survival
```

```
## Old packages: 'askpass', 'BiocGenerics', 'BiocManager', 'bit', 'bit64', 'blob',
##   'broom', 'bslib', 'cachem', 'callr', 'caTools', 'cellranger', 'cli', 'clipr',
##   'conflicted', 'cpp11', 'crayon', 'curl', 'data.table', 'DBI', 'dbplyr',
##   'digest', 'dplyr', 'dtplyr', 'evaluate', 'farver', 'fastmap', 'filelock',
##   'fontawesome', 'forcats', 'formatR', 'fs', 'futile.logger', 'gargle',
##   'generics', 'ggplot2', 'ggseqlogo', 'glue', 'googledrive', 'googlesheets4',
##   'gtable', 'gtools', 'haven', 'highr', 'hms', 'htmltools', 'httr', 'httr2',
##   'ids', 'isoband', 'jquerylib', 'jsonlite', 'lambda.r', 'lifecycle', 'locfit',
##   'lubridate', 'magrittr', 'matrixStats', 'memoise', 'modelr', 'openssl',
##   'openxlsx', 'pillar', 'pkgconfig', 'prettyunits', 'processx', 'progress',
##   'ps', 'purrr', 'R6', 'ragg', 'rappdirs', 'Rcpp', 'RcppArmadillo', 'readr',
##   'readxl', 'rematch', 'rematch2', 'reprex', 'restfulr', 'rlang', 'RSQLite',
##   'rstudioapi', 'rvest', 'S4Vectors', 'S7', 'sass', 'scales', 'selectr',
##   'stringr', 'sys', 'systemfonts', 'textshaping', 'TFMPvalue', 'tibble',
##   'tidyr', 'tidyselect', 'tidyverse', 'timechange', 'tzdb', 'utf8', 'vctrs',
##   'viridisLite', 'vroom', 'withr', 'xfun', 'xml2', 'yaml', 'zip'
```

```r
BiocManager::install("JASPAR2022")
```

```
## 'getOption("repos")' replaces Bioconductor standard repositories, see
## 'help("repositories", package = "BiocManager")' for details.
## Replacement repositories:
##     CRAN: https://cloud.r-project.org

## Bioconductor version 3.23 (BiocManager 1.30.27), R 4.6.0 (2026-04-24 ucrt)

## Warning: package(s) not installed when version(s) same as or greater than current; use
##   `force = TRUE` to re-install: 'JASPAR2022'
## Warning: Some listed binary packages have no source

## Installation paths not writeable, unable to update packages
##   path: C:/Program Files/R/R-4.6.0/library
##   packages:
##     boot, class, cluster, KernSmooth, lattice, MASS, Matrix, mgcv, nlme, nnet,
##     rpart, spatial, survival

## Old packages: 'askpass', 'BiocGenerics', 'BiocManager', 'bit', 'bit64', 'blob',
##   'broom', 'bslib', 'cachem', 'callr', 'caTools', 'cellranger', 'cli', 'clipr',
##   'conflicted', 'cpp11', 'crayon', 'curl', 'data.table', 'DBI', 'dbplyr',
##   'digest', 'dplyr', 'dtplyr', 'evaluate', 'farver', 'fastmap', 'filelock',
##   'fontawesome', 'forcats', 'formatR', 'fs', 'futile.logger', 'gargle',
##   'generics', 'ggplot2', 'ggseqlogo', 'glue', 'googledrive', 'googlesheets4',
##   'gtable', 'gtools', 'haven', 'highr', 'hms', 'htmltools', 'httr', 'httr2',
##   'ids', 'isoband', 'jquerylib', 'jsonlite', 'lambda.r', 'lifecycle', 'locfit',
##   'lubridate', 'magrittr', 'matrixStats', 'memoise', 'modelr', 'openssl',
##   'openxlsx', 'pillar', 'pkgconfig', 'prettyunits', 'processx', 'progress',
##   'ps', 'purrr', 'R6', 'ragg', 'rappdirs', 'Rcpp', 'RcppArmadillo', 'readr',
##   'readxl', 'rematch', 'rematch2', 'reprex', 'restfulr', 'rlang', 'RSQLite',
##   'rstudioapi', 'rvest', 'S4Vectors', 'S7', 'sass', 'scales', 'selectr',
##   'stringr', 'sys', 'systemfonts', 'textshaping', 'TFMPvalue', 'tibble',
##   'tidyr', 'tidyselect', 'tidyverse', 'timechange', 'tzdb', 'utf8', 'vctrs',
##   'viridisLite', 'vroom', 'withr', 'xfun', 'xml2', 'yaml', 'zip'
```

```r
library(rmarkdown)
library(knitr)
library(tinytex)
library(TFBSTools)
library(Biostrings)
```

```
## Loading required package: BiocGenerics

## Loading required package: generics

##
## Attaching package: 'generics'

## The following objects are masked from 'package:base':
##
##     as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
##     setequal, union

##
## Attaching package: 'BiocGenerics'

## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs

## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
##     order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##     rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
##     unsplit, which.max, which.min

## Loading required package: S4Vectors

## Loading required package: stats4

##
## Attaching package: 'S4Vectors'

## The following object is masked from 'package:utils':
##
##     findMatches

## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname

## Loading required package: IRanges
```

```
##
## Attaching package: 'IRanges'

## The following object is masked from 'package:grDevices':
##
##     windows

## Loading required package: XVector

## Loading required package: Seqinfo

##
## Attaching package: 'Biostrings'

## The following object is masked from 'package:base':
##
##     strsplit
```

```
library(JASPAR2022)
```

```
## Loading required package: BiocFileCache

## Loading required package: dbplyr
```

```
ddx41_promoter <- DNAString("TGCGCGCCACGGCGAAACCCCGCCTCATCCTTGCGTGAGACCCAATCCGATGTGAAGAGGCGAACGACGGCGCG

ddx41_promoter
```

```
## 2001-letter DNAString object
## seq: TGCGCGCCACGGCGAAACCCCGCCTCATCCTTGCGT...AACATAGCAAGACCCCATCTCTATAAAAAAAAATAAG
```

```
subseq_48_522_ddx41 <- subseq(ddx41_promoter, start = 48, end = 522)
subseq_48_522_ddx41
```

```
## 475-letter DNAString object
## seq: CGATGTGAAGAGGCGAACGACGGCGCGTAGGAACGA...TGCCTCCCTGACTGCGCCTGTGTGACACATACGTTC
```

```
opts <- list(species = 9606)
motifs <- getMatrixSet(JASPAR2022, opts)
pwm <- toPWM(motifs)

hits_subseq_ddx41 <- searchSeq(pwm, subseq_48_522_ddx41, seqname = "subseq_48_522_ddx41", min.score = "
hits_subseq_ddx41
```

```
## SiteSetList of length 692
## names(692): MA0030.1 MA0031.1 MA0051.1 MA0059.1 ... MA1630.2 MA1633.2 MA0597.2
```

```r
hits_ddx41_subseq_filtered_1 <- hits_subseq_ddx41[sapply(hits_subseq_ddx41, length) > 0]
hits_ddx41_subseq_filtered_1
```

```
## SiteSetList of length 21
## names(21): MA0130.1 MA0259.1 MA0671.1 MA0691.1 ... MA1986.1 MA0740.2 MA1563.2
```

```r
hits_df_ddx41 <- as(hits_ddx41_subseq_filtered_1, "data.frame")
hits_df_ddx41
```

```
##                 seqnames source feature start end  absScore  relScore strand
## 1  subseq_48_522_ddx41   TFBS    TFBS   117 122  8.083498 0.9501347      -
## 2  subseq_48_522_ddx41   TFBS    TFBS   314 321 10.037610 0.9672651      -
## 3  subseq_48_522_ddx41   TFBS    TFBS   131 139  9.023677 0.9672080      +
## 4  subseq_48_522_ddx41   TFBS    TFBS   273 281  8.646769 0.9604154      +
## 5  subseq_48_522_ddx41   TFBS    TFBS   248 257 13.691691 0.9638490      +
## 6  subseq_48_522_ddx41   TFBS    TFBS   248 257 13.769785 0.9646934      -
## 7  subseq_48_522_ddx41   TFBS    TFBS   253 260  9.173552 0.9890140      +
## 8  subseq_48_522_ddx41   TFBS    TFBS   203 210  8.390760 0.9807196      -
## 9  subseq_48_522_ddx41   TFBS    TFBS   417 424  8.176943 0.9784541      -
## 10 subseq_48_522_ddx41   TFBS    TFBS   281 289 10.589124 0.9760535      +
## 11 subseq_48_522_ddx41   TFBS    TFBS   309 317  9.798248 0.9649389      -
## 12 subseq_48_522_ddx41   TFBS    TFBS   160 175 19.337745 0.9579495      -
## 13 subseq_48_522_ddx41   TFBS    TFBS   461 467  8.003721 0.9708062      +
## 14 subseq_48_522_ddx41   TFBS    TFBS   127 136 12.348107 0.9918190      -
## 15 subseq_48_522_ddx41   TFBS    TFBS   275 285 11.606804 0.9601785      -
## 16 subseq_48_522_ddx41   TFBS    TFBS   312 322 12.422802 0.9601560      +
## 17 subseq_48_522_ddx41   TFBS    TFBS   430 440 11.742466 0.9518005      +
## 18 subseq_48_522_ddx41   TFBS    TFBS    33  41  9.295273 0.9697429      -
## 19 subseq_48_522_ddx41   TFBS    TFBS   366 373  6.716025 0.9643920      +
## 20 subseq_48_522_ddx41   TFBS    TFBS   409 416  6.309995 0.9603243      +
## 21 subseq_48_522_ddx41   TFBS    TFBS    33  40  8.782007 0.9850891      -
## 22 subseq_48_522_ddx41   TFBS    TFBS   285 292  6.226713 0.9594900      -
## 23 subseq_48_522_ddx41   TFBS    TFBS   217 224  7.695772 0.9546974      +
## 24 subseq_48_522_ddx41   TFBS    TFBS   369 378 10.071563 0.9589114      +
## 25 subseq_48_522_ddx41   TFBS    TFBS   436 447 12.644178 0.9573847      -
## 26 subseq_48_522_ddx41   TFBS    TFBS   248 257 10.053810 0.9614252      +
## 27 subseq_48_522_ddx41   TFBS    TFBS   248 257 10.052514 0.9614121      -
## 28 subseq_48_522_ddx41   TFBS    TFBS   440 449 11.498395 0.9686250      +
## 29 subseq_48_522_ddx41   TFBS    TFBS   373 382 13.497257 0.9797622      -
## 30 subseq_48_522_ddx41   TFBS    TFBS   314 322 11.009980 0.9546119      -
## 31 subseq_48_522_ddx41   TFBS    TFBS   164 171  9.547924 0.9698291      -
##          ID          TF                                  class
## 1  MA0130.1     ZNF354C                 C2H2 zinc finger factors
## 2  MA0259.1 ARNT::HIF1A    Basic helix-loop-helix factors (bHLH)
## 3  MA0671.1        NFIX     SMAD/NF-1 DNA-binding domain factors
## 4  MA0671.1        NFIX     SMAD/NF-1 DNA-binding domain factors
## 5  MA0691.1       TFAP4    Basic helix-loop-helix factors (bHLH)
## 6  MA0691.1       TFAP4    Basic helix-loop-helix factors (bHLH)
## 7  MA0719.1      RHOXF1                    Homeo domain factors
## 8  MA0719.1      RHOXF1                    Homeo domain factors
## 9  MA0719.1      RHOXF1                    Homeo domain factors
## 10 MA0738.1        HIC2                 C2H2 zinc finger factors
```

```
## 11 MA0738.1        HIC2                    C2H2 zinc finger factors
## 12 MA0088.2      ZNF143                    C2H2 zinc finger factors
## 13 MA0498.2       MEIS1                        Homeo domain factors
## 14 MA0100.3         MYB                  Tryptophan cluster factors
## 15 MA0103.3        ZEB1                        Homeo domain factors
## 16 MA1517.1        KLF6                    C2H2 zinc finger factors
## 17 MA1517.1        KLF6                    C2H2 zinc finger factors
## 18 MA1535.1       NR2C1   Nuclear receptors with C4 zinc fingers
## 19 MA1536.1       NR2C2   Nuclear receptors with C4 zinc fingers
## 20 MA1536.1       NR2C2   Nuclear receptors with C4 zinc fingers
## 21 MA1536.1       NR2C2   Nuclear receptors with C4 zinc fingers
## 22 MA1536.1       NR2C2   Nuclear receptors with C4 zinc fingers
## 23 MA0893.2        GSX2                        Homeo domain factors
## 24 MA1099.2        HES1    Basic helix-loop-helix factors (bHLH)
## 25 MA1710.1      ZNF257                    C2H2 zinc finger factors
## 26 MA1635.1     BHLHE22    Basic helix-loop-helix factors (bHLH)
## 27 MA1635.1     BHLHE22    Basic helix-loop-helix factors (bHLH)
## 28 MA1965.1         SP5                    C2H2 zinc finger factors
## 29 MA1986.1      ZNF692                    C2H2 zinc finger factors
## 30 MA0740.2       KLF14                    C2H2 zinc finger factors
## 31 MA1563.2       SOX18 High-mobility group (HMG) domain factors
##            siteSeqs
## 1            AACCAC
## 2          GGGCGTGC
## 3         GTTGCCAGG
## 4         CCTGCCAGG
## 5        GACAGCTGAT
## 6        ATCAGCTGTC
## 7          CTGATCCG
## 8          TTGAGCCT
## 9          TTTATCCC
## 10        GTGCCCAAC
## 11        GTGCCCAAG
## 12 GTCCCACAATGCCTTG
## 13          GTGACAC
## 14       GGCAACTGTC
## 15      GGCACCTGGCA
## 16      GGGCACGCCCC
## 17      CGACACACCCT
## 18        CGAGGTCGT
## 19         GAGTTCGC
## 20         GGGGTTGC
## 21         GAGGTCGT
## 22         GAGGTTGG
## 23         CCCATTGG
## 24       TTCGCGTGGG
## 25     GGGAGGCAGGGT
## 26       GACAGCTGAT
## 27       ATCAGCTGTC
## 28       TGCCTCCCTG
## 29       AGGGCCCACG
## 30        GGGGCGTGC
## 31         CACAATGC
```